

Department of Computer Engineering
Academic Term: July-November 2022

Class : B.E. Computer
Semester : VII

Subject Name :ML
Subject Code : CSC701

Practical No:	3
Title.	To Study and implement Logistic regression
Date of Performance.	01-08-2025
Roll No:	9913
Name of the Student:	Mark Lopes

Evaluation

Indicator	Very Poor	Poor	Average	Good	Excellent
Timeline (2)	More than three sessions late (0)	More than two sessions late (0.5)	Two sessions late (1)	One session late (1.5)	Early or ontime (2)
Efforts(4)	N/A	N/A	Not Completed (1)	Partially Completed (2)	Completed(4)
Legibility(2)	N/A	N/A	Poor(1)	Good(1.5)	Very Good(2)
Oral Assessment (2)	N/A	N/A	N/A	Partially Understood the concept (1)	Understood the concept(2)

```
In [ ]: import numpy as np
import pandas as pd

from sklearn import preprocessing
import matplotlib.pyplot as plt
plt.rc("font", size=14)
import seaborn as sns
sns.set(style="white") #white background style for seaborn plots
sns.set(style="whitegrid", color_codes=True)

import warnings
warnings.simplefilter(action='ignore')
```

New Section

```
In [ ]: train_df = pd.read_csv("/content/titanic.csv")

# Read CSV test data file into DataFrame
test_df = pd.read_csv("/content/titanic.csv")

# preview train data
train_df.head()
```

```
Out[ ]: 
```

	passengerid	pclass	survived	name	sex	age	sibsp	parch	ticket	fare	cabin	embarked
0	1	1	1	Allen, Miss. Elisabeth Walton	female	29.0000	0	0	24160	211.3375	B5	S
1	2	1	1	Allison, Master. Hudson Trevor	male	0.9167	1	2	113781	151.5500	C22 C26	S
2	3	1	0	Allison, Miss. Helen Loraine	female	2.0000	1	2	113781	151.5500	C22 C26	S
3	4	1	0	Allison, Mr. Hudson Joshua Creighton	male	30.0000	1	2	113781	151.5500	C22 C26	S
4	5	1	0	Allison, Mrs. Hudson J C (Bessie Waldo Daniels)	female	25.0000	1	2	113781	151.5500	C22 C26	S

```
In [ ]: print('The number of samples into the train data is {}'.format(train_df.shape[0]))

The number of samples into the train data is 1309.
```

```
In [ ]: test_df.head()
```

```
Out[ ]: 
```

	passengerid	pclass	survived	name	sex	age	sibsp	parch	ticket	fare	cabin	embarked
0	1	1	1	Allen, Miss. Elisabeth Walton	female	29.0000	0	0	24160	211.3375	B5	S
1	2	1	1	Allison, Master. Hudson Trevor	male	0.9167	1	2	113781	151.5500	C22 C26	S
2	3	1	0	Allison, Miss. Helen Loraine	female	2.0000	1	2	113781	151.5500	C22 C26	S
3	4	1	0	Allison, Mr. Hudson Joshua Creighton	male	30.0000	1	2	113781	151.5500	C22 C26	S
4	5	1	0	Allison, Mrs. Hudson J C (Bessie Waldo Daniels)	female	25.0000	1	2	113781	151.5500	C22 C26	S

```
In [ ]: print('The number of samples into the test data is {}'.format(test_df.shape[0]))

The number of samples into the test data is 1309.
```

```
In [ ]: train_df.isnull().sum()
```

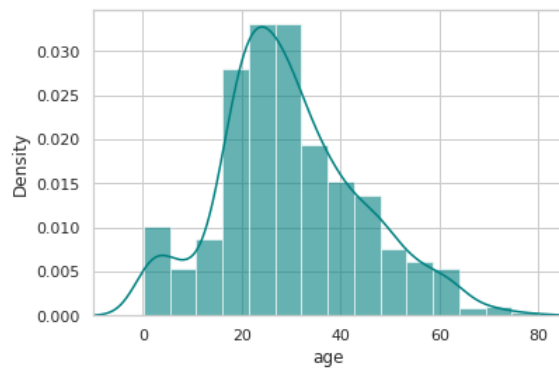
```
Out[ ]: passengerid      0
pclass              0
survived            0
name                0
sex                 0
age                263
sibsp               0
parch              0
ticket              0
fare                1
cabin             1014
embarked            2
dtype: int64
```

```
In [ ]: print('Percent of missing "Age" records is %.2f%%' % ((train_df['age'].isnull().sum()/train_df.shape[0])*100))

Percent of missing "Age" records is 20.09%
```

```
In [ ]: ax = train_df["age"].hist(bins=15, density=True, stacked=True, color='teal', alpha=0.6)
train_df["age"].plot(kind='density', color='teal')
ax.set(xlabel='age')
```

```
plt.xlim(-10,85)
plt.show()
```



```
In [ ]: print('The mean of "Age" is %.2f' %(train_df["age"].mean(skipna=True)))
# median age
print('The median of "Age" is %.2f' %(train_df["age"].median(skipna=True)))
```

The mean of "Age" is 29.88
The median of "Age" is 28.00

```
In [ ]: print('Percent of missing "Cabin" records is %.2f%%' %((train_df['cabin'].isnull().sum()/train_df.shape[0])*100))
```

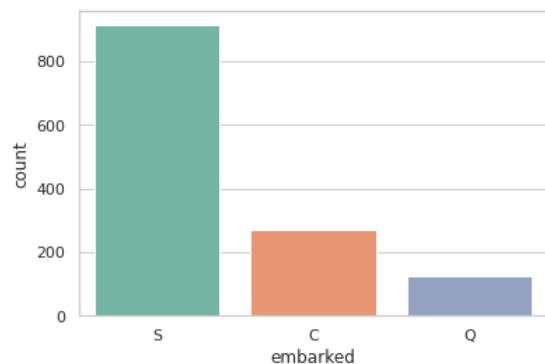
Percent of missing "Cabin" records is 77.46%

```
In [ ]: print('Percent of missing "Embarked" records is %.2f%%' %((train_df['embarked'].isnull().sum()/train_df.shape[0])*100))
```

Percent of missing "Embarked" records is 0.15%

```
In [ ]: print('Boarded passengers grouped by port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton):')
print(train_df['embarked'].value_counts())
sns.countplot(x='embarked', data=train_df, palette='Set2')
plt.show()
```

Boarded passengers grouped by port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton):
S 914
C 270
Q 123
Name: embarked, dtype: int64



```
In [ ]: print('The most common boarding port of embarkation is %s.' %train_df['embarked'].value_counts().idxmax())
```

The most common boarding port of embarkation is S.

```
In [ ]: train_data = train_df.copy()
train_data["age"].fillna(train_df["age"].median(skipna=True), inplace=True)
train_data["embarked"].fillna(train_df["embarked"].value_counts().idxmax(), inplace=True)
train_data.drop('cabin', axis=1, inplace=True)
```

```
In [ ]: train_data.isnull().sum()
```

```
Out[ ]: passengerid    0
pclass            0
survived          0
name              0
sex              0
age              0
sibsp            0
parch            0
ticket           0
fare             1
embarked         0
dtype: int64
```

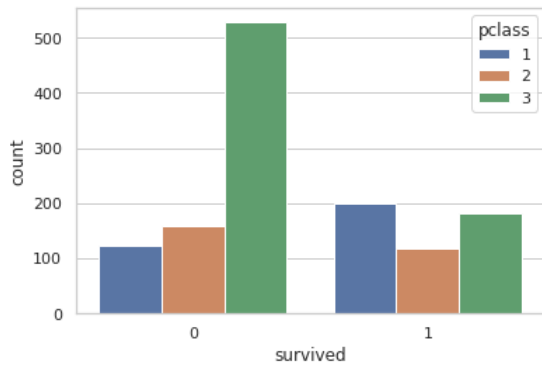
```
In [ ]: train_data.head()
```

```
Out [ ]:
```

	passengerid	pclass	survived	name	sex	age	sibsp	parch	ticket	fare	embarked
0	1	1	1	Allen, Miss. Elisabeth Walton	female	29.0000	0	0	24160	211.3375	S
1	2	1	1	Allison, Master. Hudson Trevor	male	0.9167	1	2	113781	151.5500	S
2	3	1	0	Allison, Miss. Helen Loraine	female	2.0000	1	2	113781	151.5500	S
3	4	1	0	Allison, Mr. Hudson Joshua Creighton	male	30.0000	1	2	113781	151.5500	S
4	5	1	0	Allison, Mrs. Hudson J C (Bessie Waldo Daniels)	female	25.0000	1	2	113781	151.5500	S

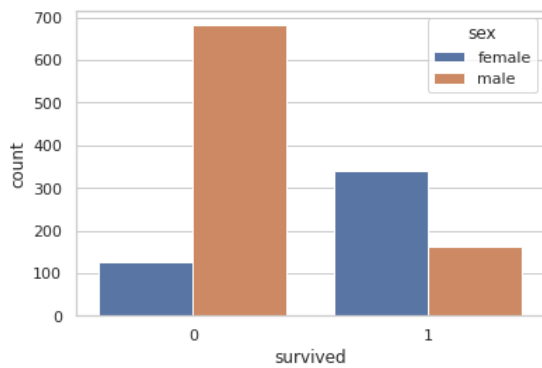
```
In [ ]: sns.countplot(x='survived', hue='pclass', data=train_df)
```

```
Out [ ]: <matplotlib.axes._subplots.AxesSubplot at 0x7ffac6809a10>
```



```
In [ ]: sns.countplot(x='survived', hue='sex', data=train_df)
```

```
Out [ ]: <matplotlib.axes._subplots.AxesSubplot at 0x7ffac67084d0>
```



```
In [ ]: def add_age(cols):
    Age = cols[0]
    Pclass = cols[1]
    if pd.isnull(Age):
        return int(train_df[train_df["pclass"] == Pclass]["age"].mean())
    else:
        return Age
```

```
In [ ]: train_df["age"] = train_df[["age", "pclass"]].apply(add_age,axis=1)
```

```
In [ ]: train_df.drop("cabin",inplace=True,axis=1)
```

```
In [ ]: train_df.dropna(inplace=True)
```

```
In [ ]: pd.get_dummies(train_df["sex"])
```

```
Out[ ]:
```

	female	male
0	1	0
1	0	1
2	1	0
3	0	1
4	1	0
...	...	...
1304	1	0
1305	1	0
1306	0	1
1307	0	1
1308	0	1

1306 rows × 2 columns

```
In [ ]: sex = pd.get_dummies(train_df["sex"],drop_first=True)
```

```
In [ ]: embarked = pd.get_dummies(train_df["embarked"],drop_first=True)
pclass = pd.get_dummies(train_df["pclass"],drop_first=True)
```

```
In [ ]: train = pd.concat([train_df,pclass,sex,embarked],axis=1)
```

```
In [ ]: train.drop(["passengerid","pclass","name","sex","ticket","embarked"],axis=1,inplace=True)
```

```
In [ ]: X = train.drop("survived",axis=1)
y = train["survived"]
```

```
In [ ]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3, random_state = 101)
```

```
In [ ]: from sklearn.linear_model import LogisticRegression
logmodel = LogisticRegression()
logmodel.fit(X_train,y_train)
```

```
Out[ ]: LogisticRegression()
```

```
In [ ]: predictions = logmodel.predict(X_test)
from sklearn.metrics import classification_report
print(classification_report(y_test, predictions))
```

	precision	recall	f1-score	support
0	0.84	0.86	0.85	247
1	0.75	0.73	0.74	145
accuracy			0.81	392
macro avg	0.80	0.79	0.80	392
weighted avg	0.81	0.81	0.81	392

```
In [ ]: from sklearn.metrics import confusion_matrix
confusion_matrix(y_test, predictions)
```

```
Out[ ]: array([[212, 35],
               [ 39, 106]])
```

```
In [ ]: %matplotlib inline
from sklearn.metrics import confusion_matrix
import itertools
import matplotlib.pyplot as plt
```

```
In [ ]: def plot_confusion_matrix(cm, classes,
                                normalize=False,
                                title='Confusion matrix',
                                cmap=plt.cm.Blues):
    """
    This function prints and plots the confusion matrix.
    Normalization can be applied by setting `normalize=True`.
    """
    plt.imshow(cm, interpolation='nearest', cmap=cmap)
    plt.title(title)
    plt.colorbar()
    tick_marks = np.arange(len(classes))
    plt.xticks(tick_marks, classes, rotation=45)
    plt.yticks(tick_marks, classes)

    if normalize:
        cm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
        print("Normalized confusion matrix")
```

```

else:
    print('Confusion matrix, without normalization')

print(cm)

thresh = cm.max() / 2.
for i, j in itertools.product(range(cm.shape[0]), range(cm.shape[1])):
    plt.text(j, i, cm[i, j],
             horizontalalignment="center",
             color="white" if cm[i, j] > thresh else "black")

plt.tight_layout()
plt.ylabel('True label')
plt.xlabel('Predicted label')

```

```

In [ ]: cm = confusion_matrix(y_true=y_test, y_pred=predictions)
cm_plot_labels = ["True", "False"]
plot_confusion_matrix(cm=cm, classes=cm_plot_labels, title='Confusion Matrix')

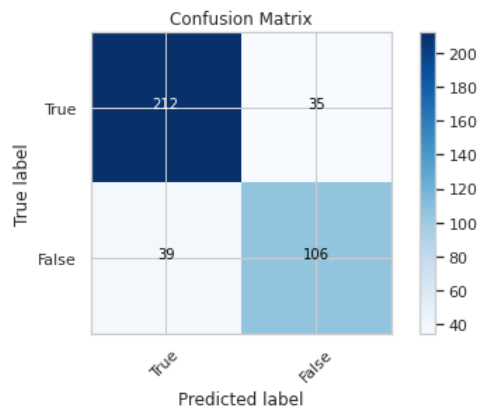
```

Confusion matrix, without normalization

```

[[212  35]
 [ 39 106]]

```



Q What is a logistic function?
What is a logistic function's range of values?

Ans A logistic function is a type of sigmoid curve that models growth which starts exponentially but then levels off as it approaches a maximum limit.
The values of a logistic function will range from 0 to 1. The importance of x will vary from $-\infty$ to ∞ .

Q What makes logistic regression so well-liked?

Logistic regression is famous because it can convert the values of logits (log odds), which can range from $-\infty$ to ∞ to a range between 0 and 1. As logistic function outputs the probability of occurrence of an event, it can be applied to many real-life scenarios. It is for this reason that the logistic regression model is prevalent.

Q How can a logistic regression model's probability be stated as a conditional probability?

Ans $P(\text{discrete value of target variable } X_1, X_2, X_3 \dots X_n)$. it is the probability of the target variable taking up a discrete value (either 0 or 1 in case of binary classification problems) when the values of independent variables are given. For example, the probability an employee will resign (target variable) given his attributes such as his age, salary, KPIs, etc.

Q What are the outputs of the logistic model and the logistic function?

Ans The logistic model outputs the logit, which refers to the log odds, while the logistic function transforms this output into probabilities. The logistic model calculates a weighted sum of the input features, and this result (the logit) is then passed through the logistic function to obtain a probability value between 0 and 1.